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← Placement, section 001, Summer 2 2026

PLAYGROUND 5 - Pooled questions (Test)

INSTRUCTOR

Alberto Gandolfi

New York University Abu Dhabi, United Arab Emirate

Due Date: SAT, JUN 13, 2026 4:00 PM GMT+4

REQUEST EXTENSION

Current Score: – / 100 POINTS | 0.0 %

Scoring and Assignment Information

QUESTION	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
POINTS	– / 6	– / 6	– / 6	– / 6	– / 6	– / 6	– / 6	– / 6	– / 5	– / 5	– / 6	– / 6	– / 6	– / 6	– / 6	– / 6	– / 6

Assignment Submission

For this assignment, you submit answers by question parts. The number of submissions remaining for each question part only changes if you submit or change the answer.

Assignment Scoring

Your last submission is used for your score.

ASK YOUR TEACHER

2. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 1.2.052.

Evaluate each expression, without using a calculator. (Simplify your answers completely.)

(a) $64^{1/3}$

(b) $(-27)^{1/3}$

(c) $-\left(\frac{1}{27}\right)^{1/3}$

3. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 1.3.049.

Perform the indicated operations and simplify.

$$(6x - 9)(x^2 - x + 1)$$

4. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 1.4.028.

Perform the multiplication or division and simplify.

$$\frac{x^2 - 81}{x^2 - 25} \cdot \frac{x + 5}{x + 9}$$

5. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 1.5.080.

Find all real solutions of the equation. (Enter your answers as a comma-separated list. If there is no real solution, enter NO REAL SOLUTION.)

$$\frac{9}{x - 9} - \frac{18}{x^2} = 0$$

x =

6. [– / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 1.8.039.

Solve the nonlinear inequality. Express the solution using interval notation.

$x(2x + 5) \geq 0$

Graph the solution set.

Use the tools to enter your answer.



NO SOLUTION

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7. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 1.10.024.

Find an equation of the line that satisfies the given conditions.

Slope $\frac{6}{7}$; y-intercept 5

8. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 2.4.513.XP.

A function is given.

$$f(z) = 5 - 4z^2; \quad z = -2, z = 0$$

(a) Determine the net change between the given values of the variable.

(b) Determine the average rate of change between the given values of the variable.

9. [- / 5 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

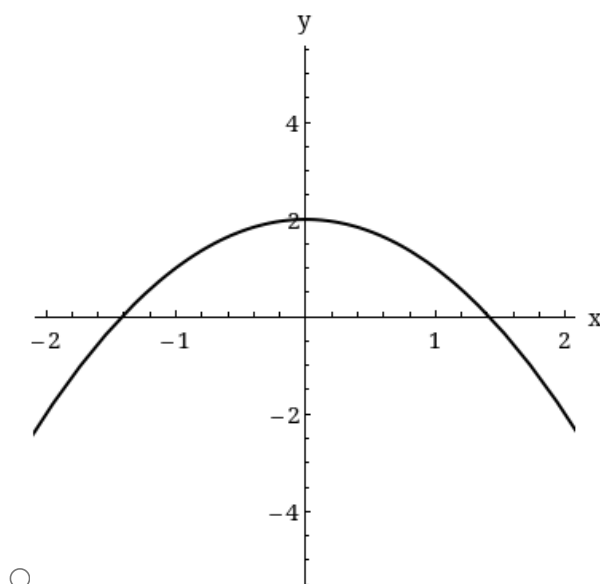
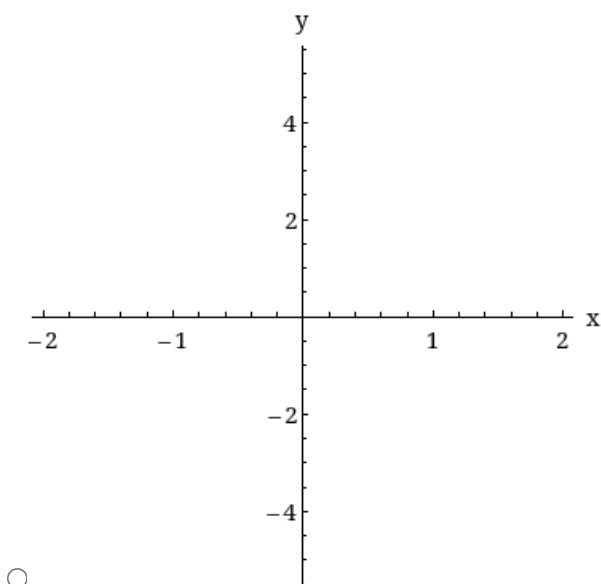
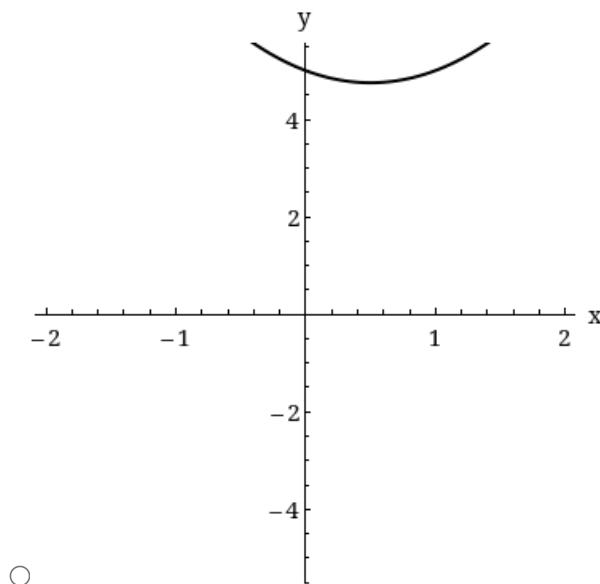
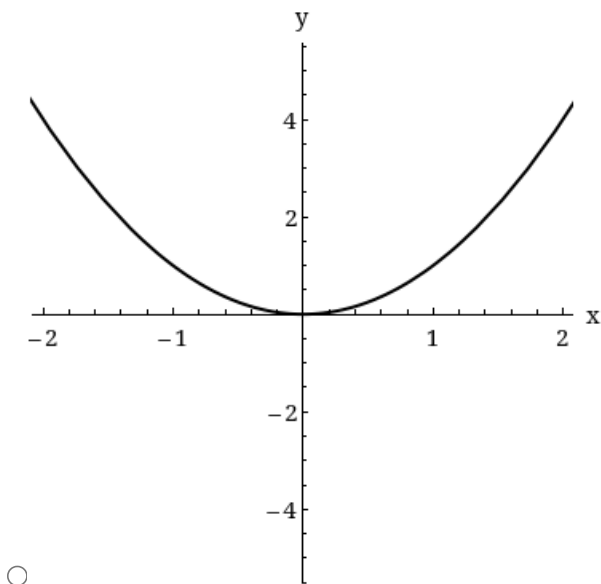
SPreCalc8M 2.2.527.XP.

A graphing calculator is recommended.

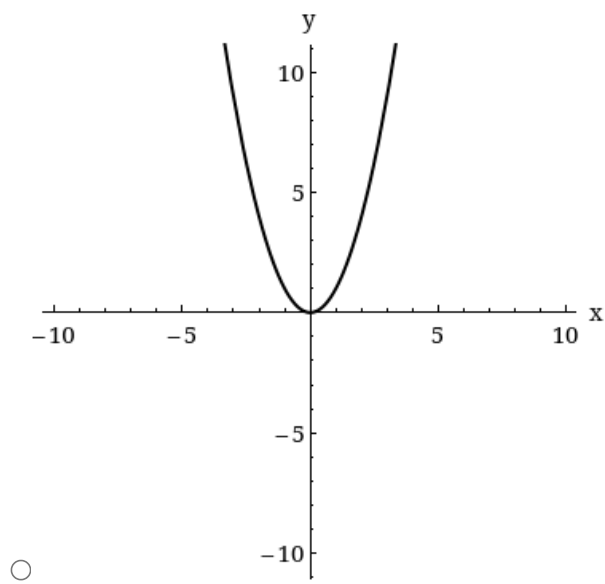
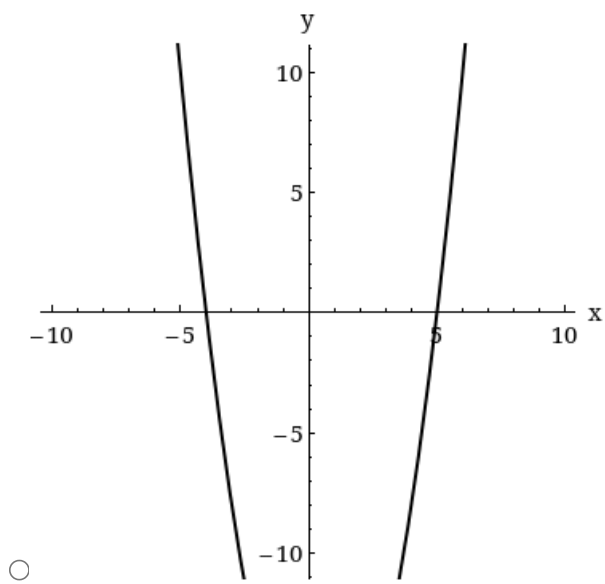
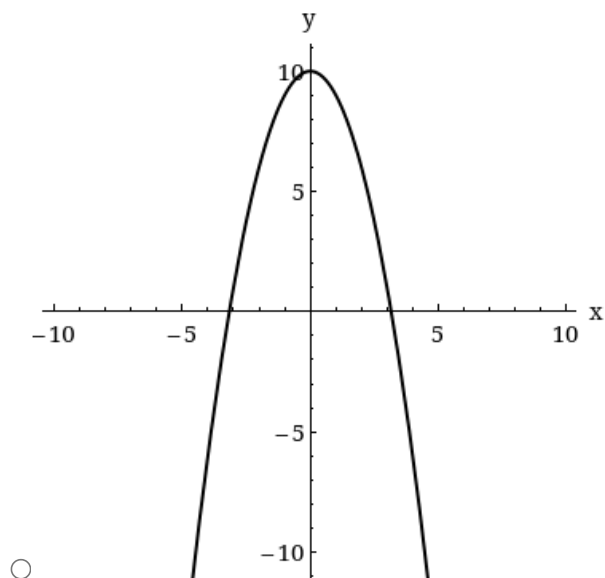
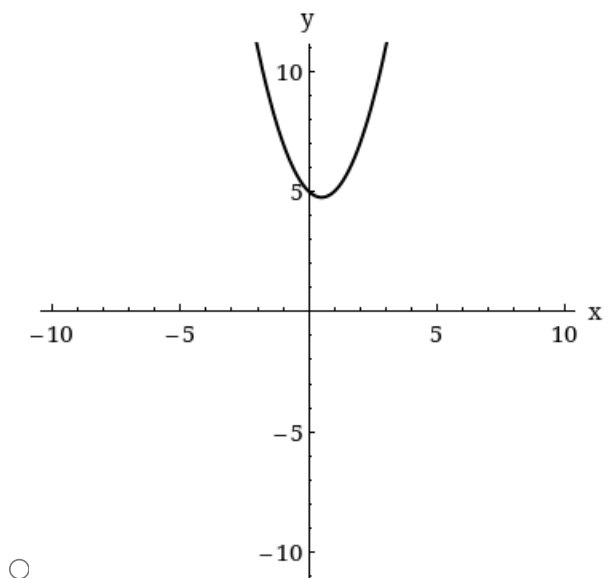
Graph the function in each of the given viewing rectangles.

$$g(x) = x^2 - x - 20$$

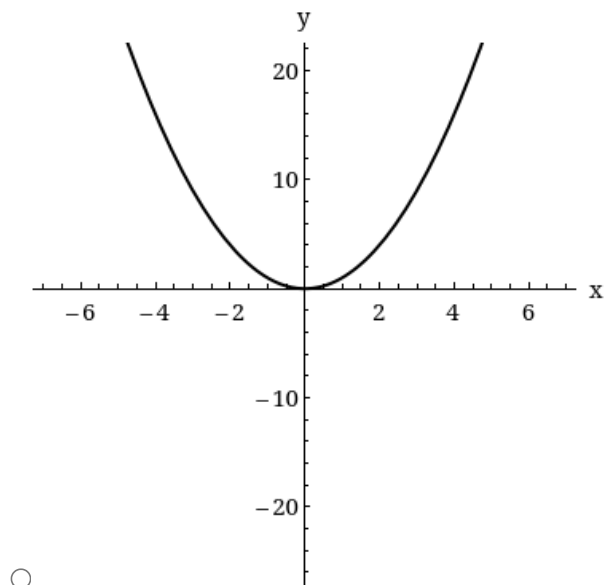
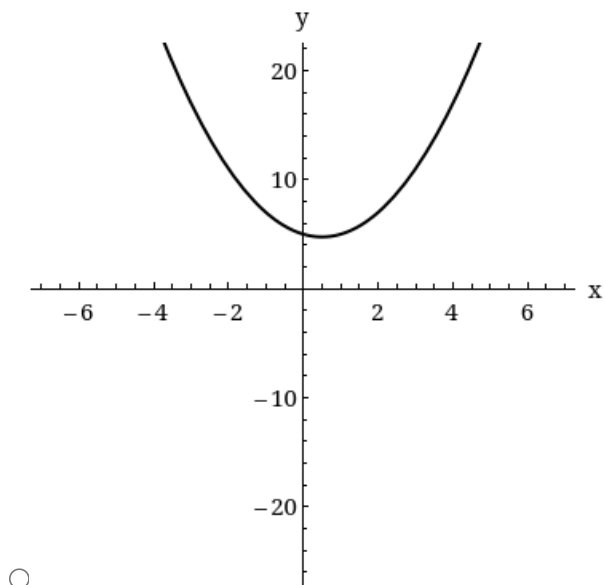
(a) $[-2, 2]$ by $[-5, 5]$

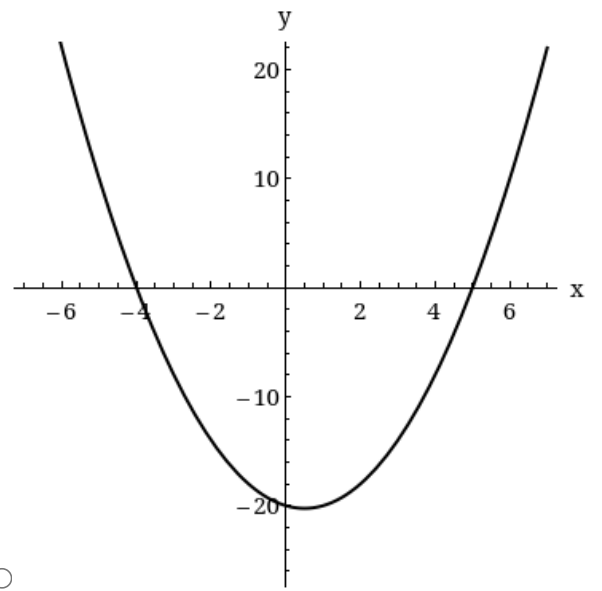
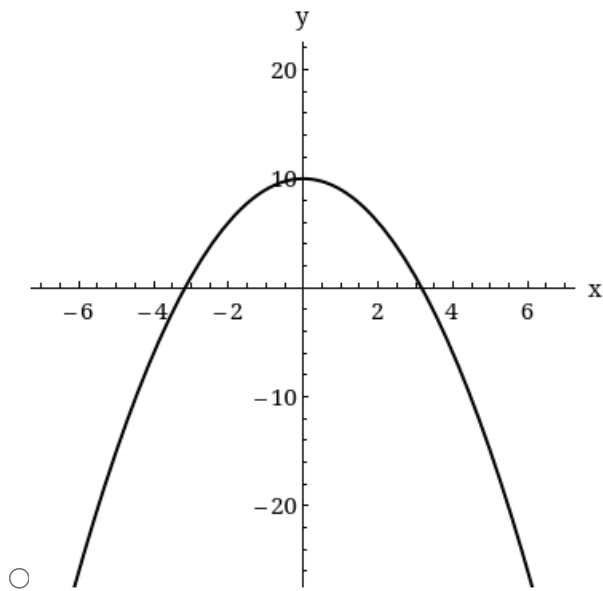


(b) $[-10, 10]$ by $[-10, 10]$

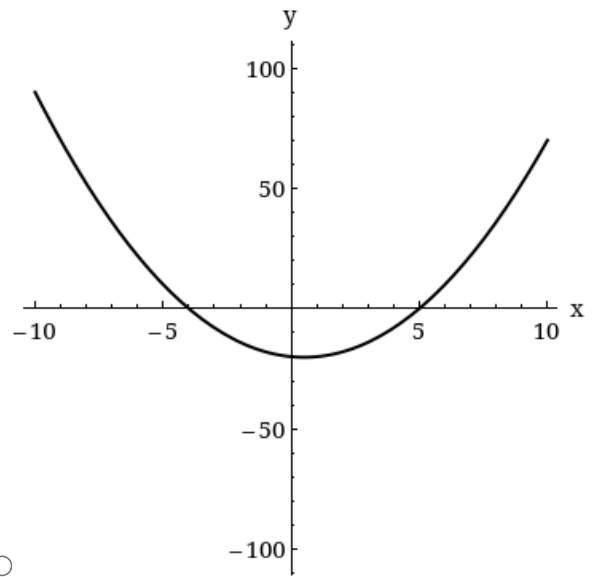
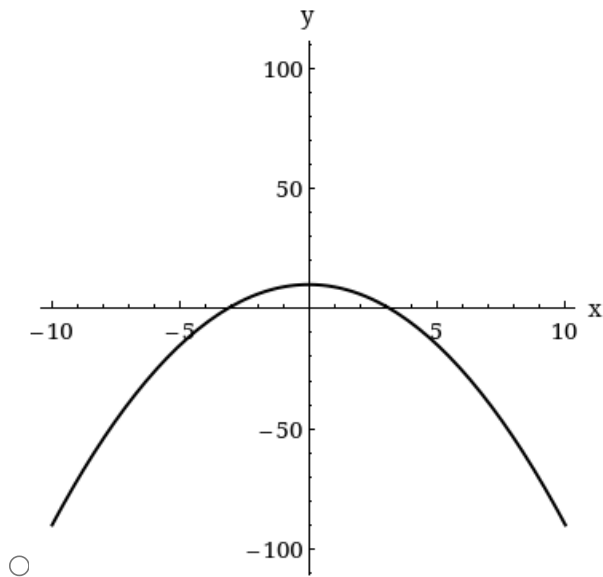


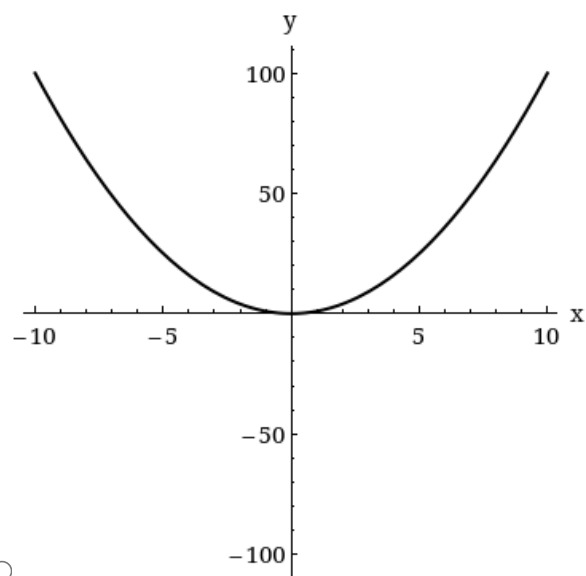
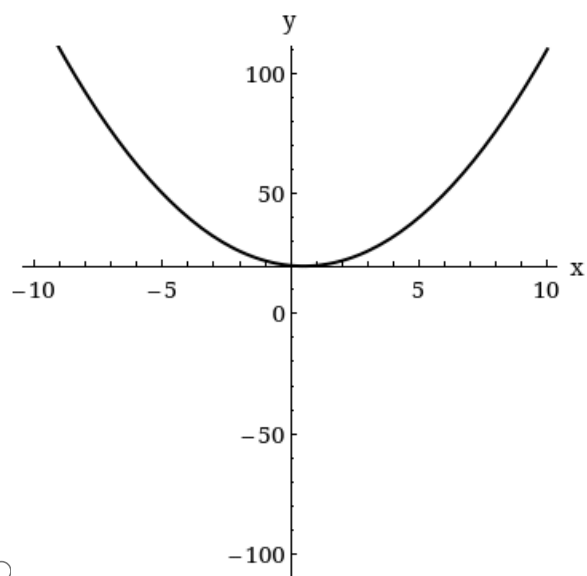
(c) $[-7, 7]$ by $[-25, 20]$





(d) $[-10, 10]$ by $[-100, 100]$





Select the viewing window that produces the most appropriate graph of the function.

- ☐ $[-2, 2]$ by $[-5, 5]$
- ☐ $[-10, 10]$ by $[-10, 10]$
- ☐ $[-7, 7]$ by $[-25, 20]$
- ☐ $[-10, 10]$ by $[-100, 100]$

10. [- / 5 Points]

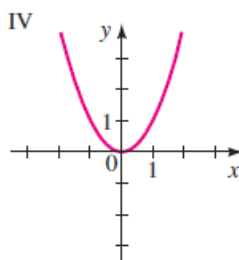
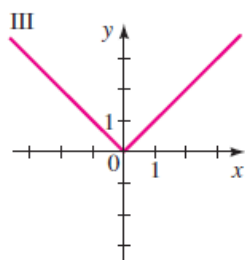
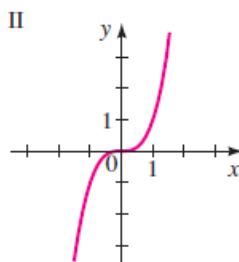
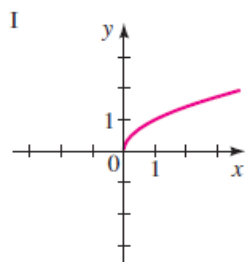
DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 2.2.004.

Match the function with its graph.



(a) $f(x) = x^2$

(b) $f(x) = x^3$

(c) $f(x) = \sqrt{x}$

(d) $f(x) = |x|$

11. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

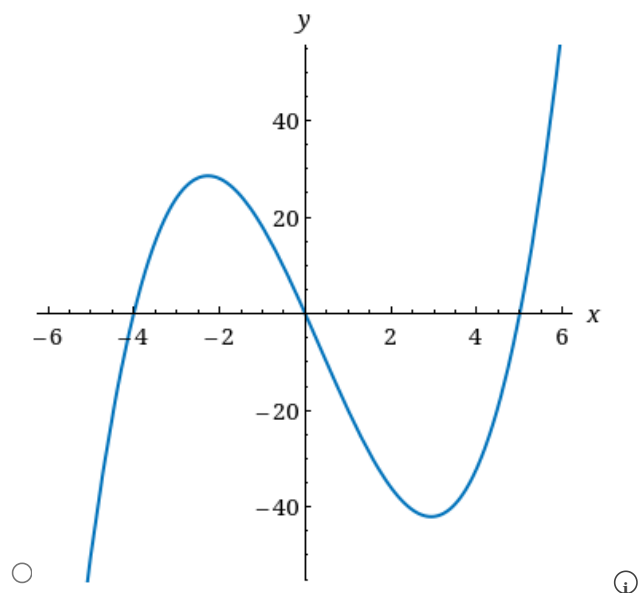
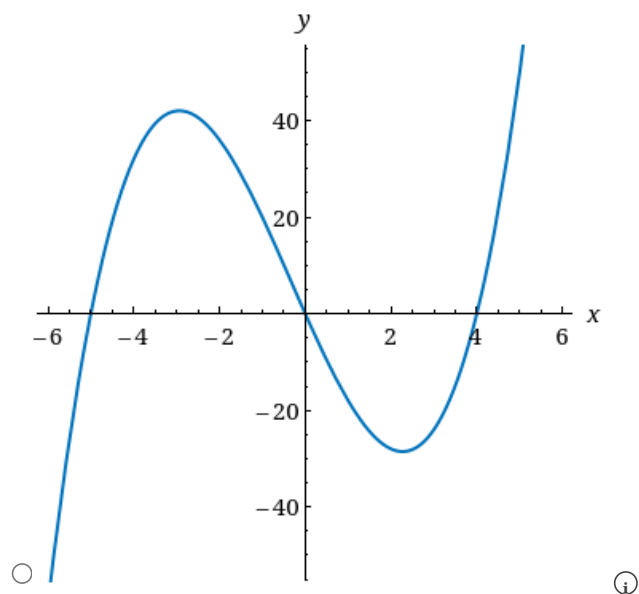
SPreCalc8M 3.2.032.

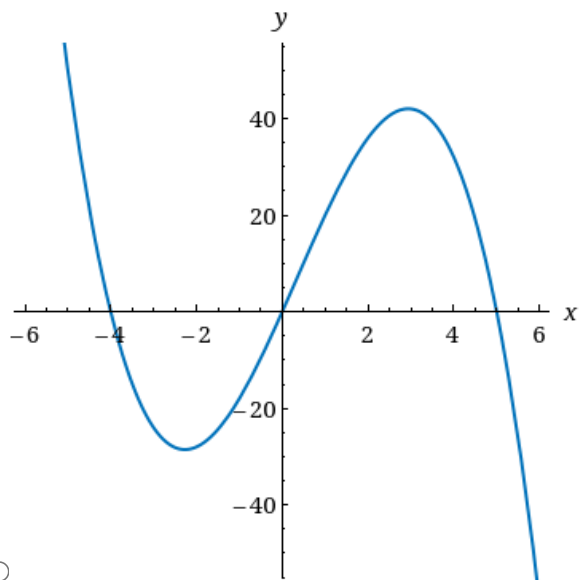
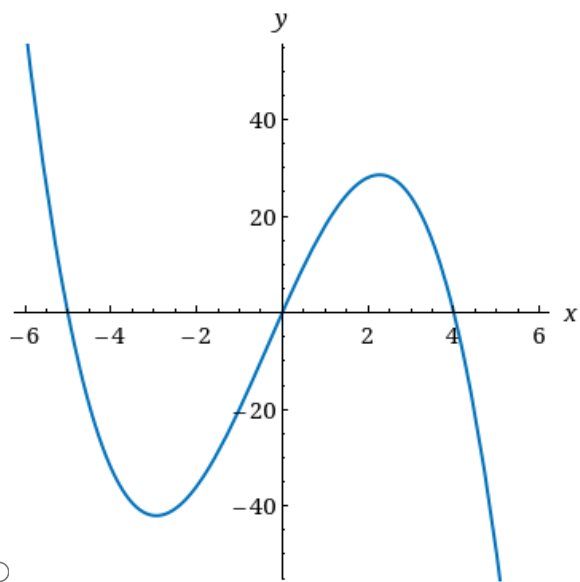
Factor the polynomial and use the factored form to find the zeros. (Enter your answers as a comma-separated list. Enter all answers using the appropriate multiplicities.)

$$P(x) = x^3 + x^2 - 20x$$

 $x =$

Sketch the graph.



☐☐☐☐

12. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 4.2.045.MI.

If \$1,000 is invested at an interest rate of 6.5% per year, compounded continuously, find the value of the investment after the given number of years. (Round your answers to the nearest cent.)

(a) 3 years

\$

(b) 6 years

\$

(c) 18 years

\$

13. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 5.2.528.XP.

The terminal point $P(x, y)$ determined by a real number t is given. Find $\sin(t)$, $\cos(t)$, and $\tan(t)$.

$$\left(-\frac{3}{5}, -\frac{4}{5}\right)$$

 $\sin(t) =$

 $\cos(t) =$

 $\tan(t) =$

14. [- / 6 Points]

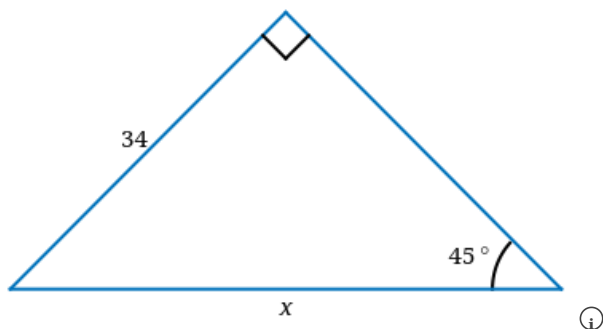
DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 6.2.018.MI.

Find the side labeled x in the following figure.

 $x =$

15. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 7.1.035.

Verify the identity using the fundamental trigonometric identities.

$$\frac{\cos(u) \sec(u)}{\tan(u)} = \cot(u)$$

Use a Reciprocal Identity to rewrite the expression, and then simplify.

$$\frac{\cos(u) \sec(u)}{\tan(u)} = (\cos(u) \sec(u)) \left(\frac{\quad}{\quad} \right)$$

$$= \cos(u) \cdot \frac{1}{\cos(u)} \cdot \left(\frac{\quad}{\quad} \right)$$

$$= \frac{\quad}{\quad}$$

16. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 8.6.028.MI.

Find the indicated quantity, given that $\mathbf{u} = 4\mathbf{i} + \mathbf{j}$, $\mathbf{v} = \mathbf{i} - 3\mathbf{j}$, and $\mathbf{w} = 3\mathbf{i} + 2\mathbf{j}$.

$$(\mathbf{u} \cdot \mathbf{v})(\mathbf{u} \cdot \mathbf{w})$$

17. [- / 6 Points]

DETAILS

MY NOTES

ASK YOUR TEACHER

SPreCalc8M 11.1.064.

Write the sum without using sigma notation.

$$\sum_{k=6}^9 k(k+3)$$

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